THE SHARPEST DIAGNOSTIC

When A is $n \times n$, the determinant decides whether the coefficient matrix is adequate; when it is not, the rank comparison with the augmented matrix decides whether the specific $\mathbf{b}$ is reachable.

SQUARE SYSTEM

$A\mathbf{x} = \mathbf{b}$, A is $n \times n$

split by $\det(A)$

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CASE 1

$\det(A) \neq 0$

Exactly one

solution for every $\mathbf{b}$

the matrix is invertible and the rank is n; existence and uniqueness both hold universally

$$\mathbf{x} = A \quad \mathbf{b}$$

∞ or 0

CASE 2

$\det(A) = 0$

Infinitely many or none

depends on $\mathbf{b}$

the matrix is singular and the rank is less than n: infinitely many solutions for $\mathbf{b}$ in the column space, none for $\mathbf{b}$ outside it

TAKEAWAY

With $\det(A) \neq 0$ the right-hand side does not matter. With $\det(A) = 0$ everything depends on whether $\mathbf{b}$ lies in the column space.